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DSA0602 Data Handling and Visulaization - Slot A

Unit II - Class Activity 1 **Date: 20.07.2026**

1) Hospital Patient Admission Analysis:

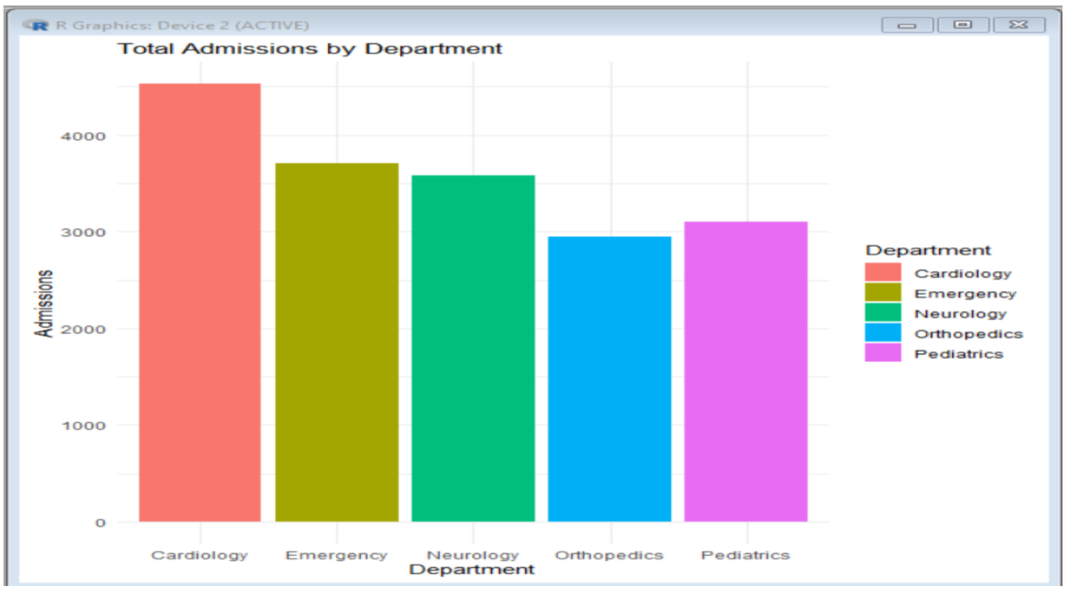
R code:

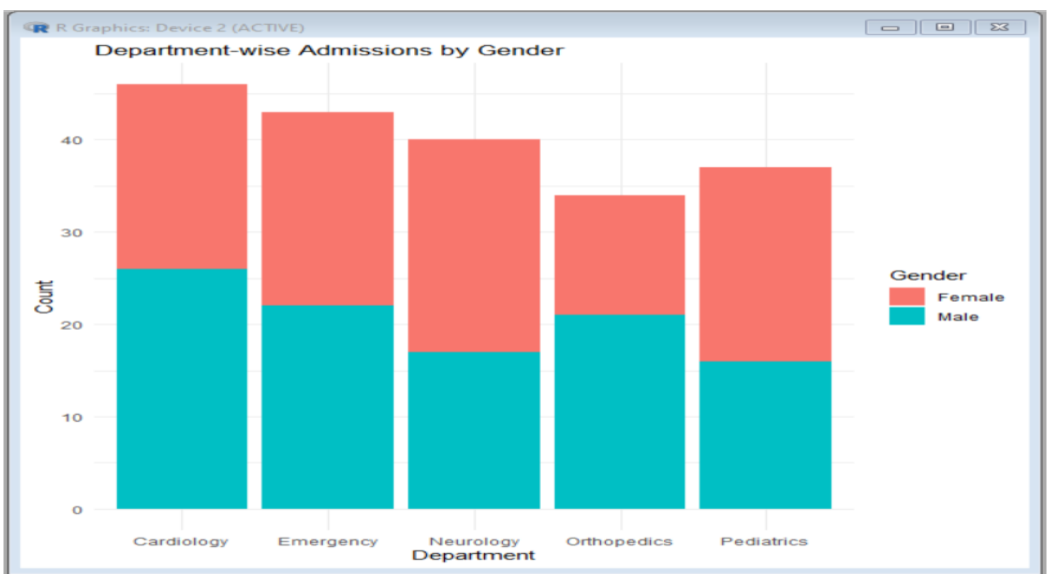
```r
# Install packages (Run once)
install.packages(c("ggplot2","dplyr","treemap"))

# Load libraries
library(ggplot2)
library(dplyr)
library(treemap)

# -----------------------------
# Create Sample Dataset
# -----------------------------
set.seed(123)

hospital <- data.frame(
  Department = sample(c("Cardiology","Neurology","Pediatrics",
                      "Orthopedics","Emergency"),
                     200, replace = TRUE),
  Admissions = sample(20:150, 200, replace = TRUE),
  Gender = sample(c("Male","Female"), 200, replace = TRUE)
)

# -----------------------------
# 1. Bar Chart
# -----------------------------
dept_count <- hospital %>%
  group_by(Department) %>%
  summarise(Total_Admissions = sum(Admissions))

ggplot(dept_count,
       aes(x = Department,
           y = Total_Admissions,
           fill = Department)) +
  geom_bar(stat = "identity") +
```

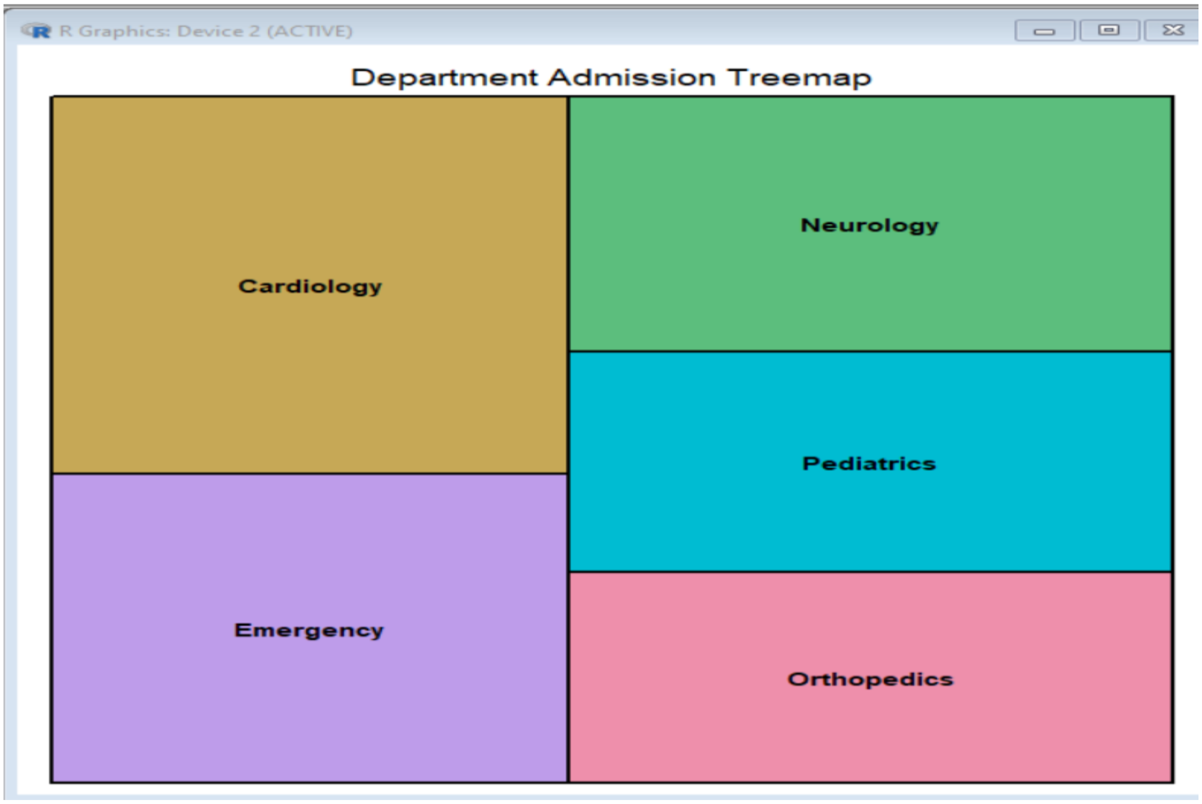
```r
  labs(title = "Total Admissions by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()

# ----------------------------
# 2. Stacked Bar Chart
# ----------------------------
ggplot(hospital,
       aes(x = Department,
           fill = Gender)) +
  geom_bar() +
  labs(title = "Department-wise Admissions by Gender",
       x = "Department",
       y = "Count") +
  theme_minimal()

# ----------------------------
# 3. Treemap
# ----------------------------
treemap(dept_count,
        index = "Department",
        vSize = "Total_Admissions",
        title = "Department Admission Treemap")

# ----------------------------
# 4. Box Plot
# ----------------------------
ggplot(hospital,
       aes(x = Department,
           y = Admissions,
           fill = Department)) +
  geom_boxplot() +
  labs(title = "Admission Variability by Department",
       x = "Department",
       y = "Admissions") +
  theme_minimal()

# ----------------------------
# 5. Histogram
# ----------------------------
ggplot(hospital,
       aes(x = Admissions)) +
  geom_histogram(binwidth = 10,
```

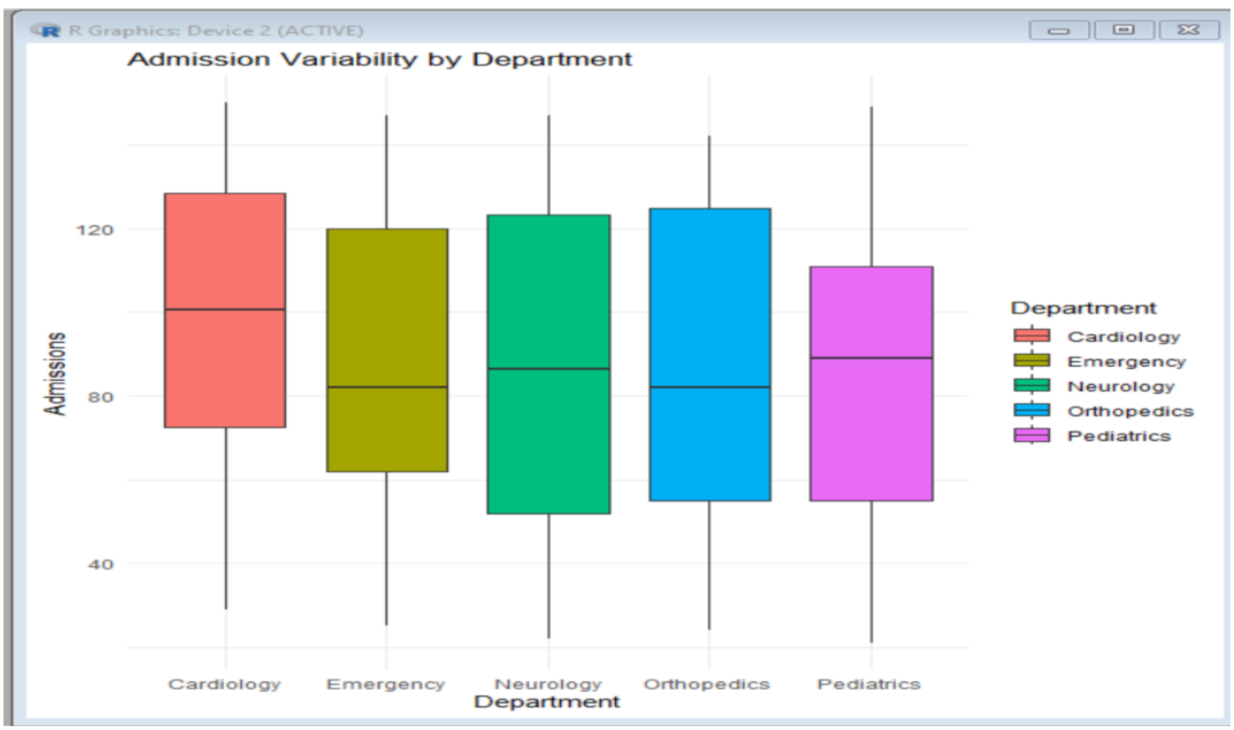
```
            fill = "skyblue",
            color = "black") +
  labs(title = "Distribution of Patient Admissions",
       x = "Admissions",
       y = "Frequency") +
  theme_minimal()
```

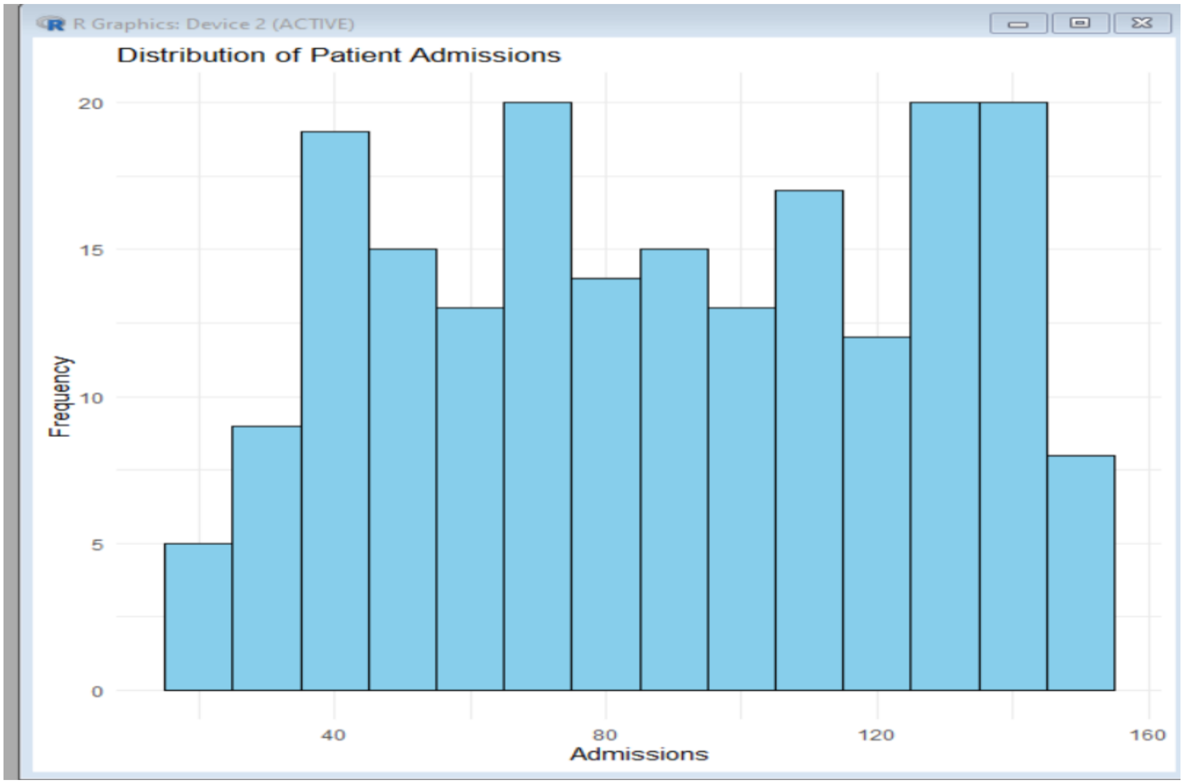
Output:






R Graphics: Device 2 (ACTIVE)
Department Admission Treemap
Cardiology
Neurology
Pediatrics
Emergency
Orthopedics


R Graphics: Device 2 (ACTIVE)
Admission Variability by Department
Admissions
120
80
40
Cardiology
Emergency
Neurology
Orthopedics
Pediatrics
Department
Department
Cardiology
Emergency
Neurology
Orthopedics
Pediatrics



2) E-Commerce Customer Spending Behaviour:

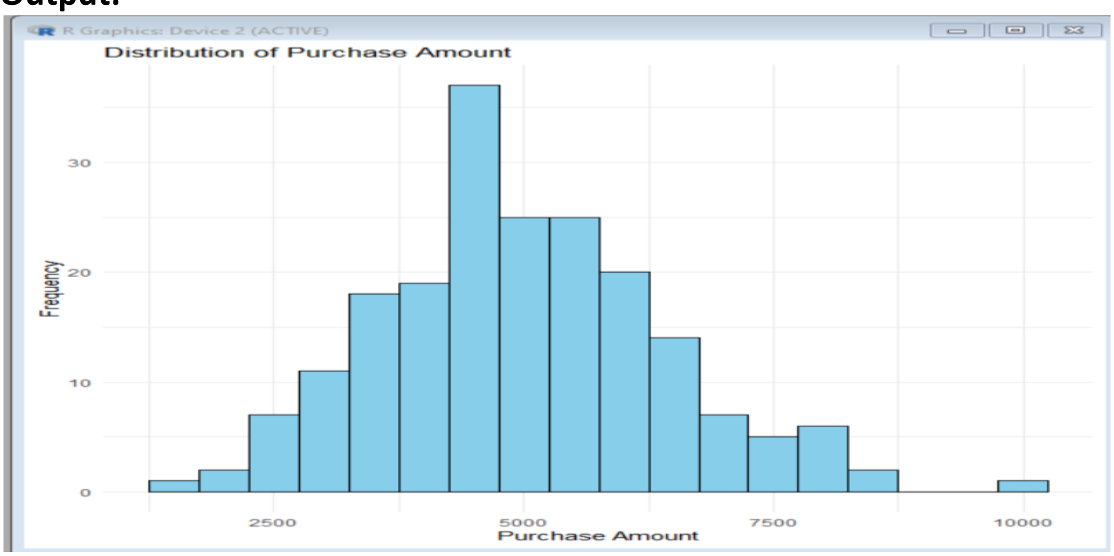
R code:

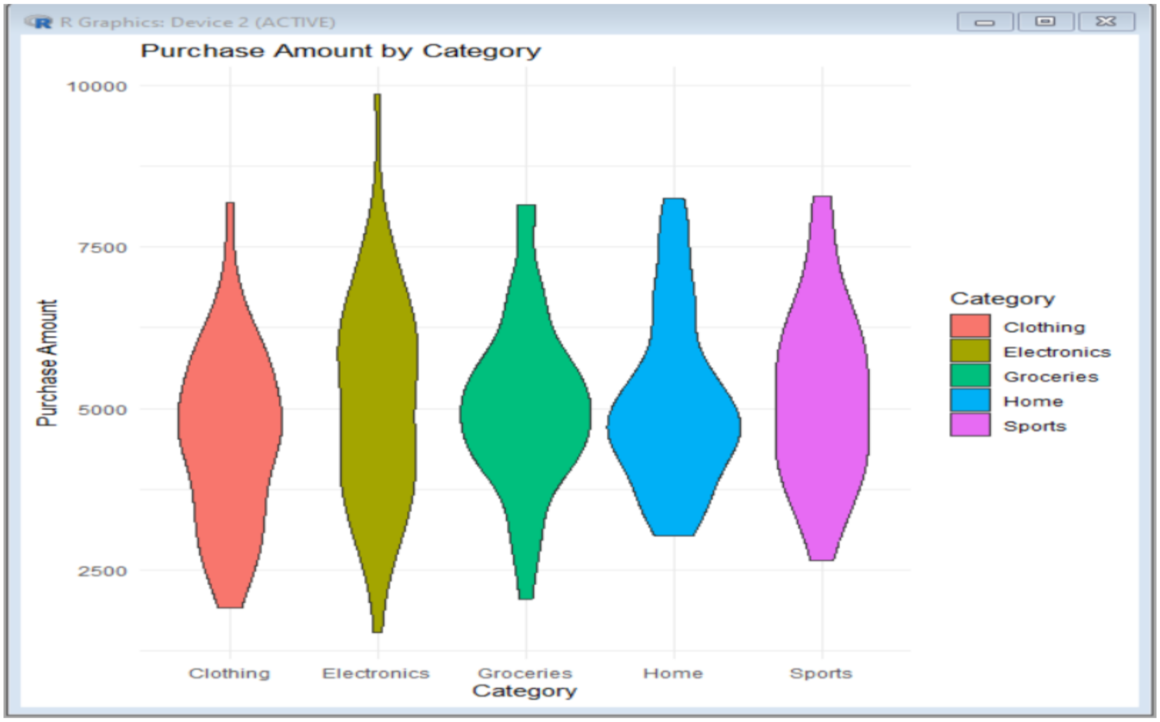
```
# Install packages (Run once)
install.packages(c("ggplot2","dplyr"))

# Load libraries
library(ggplot2)
library(dplyr)

# ----------------------------
# Create Sample Dataset
# ----------------------------
set.seed(123)

customers <- data.frame(
  CustomerID = 1:200,
  PurchaseAmount = round(rnorm(200, mean = 5000, sd = 1500)),
  OrderFrequency = sample(1:20, 200, replace = TRUE),
  Category = sample(c("Electronics","Clothing","Groceries","Home","Sports"),
                200, replace = TRUE)
)
```

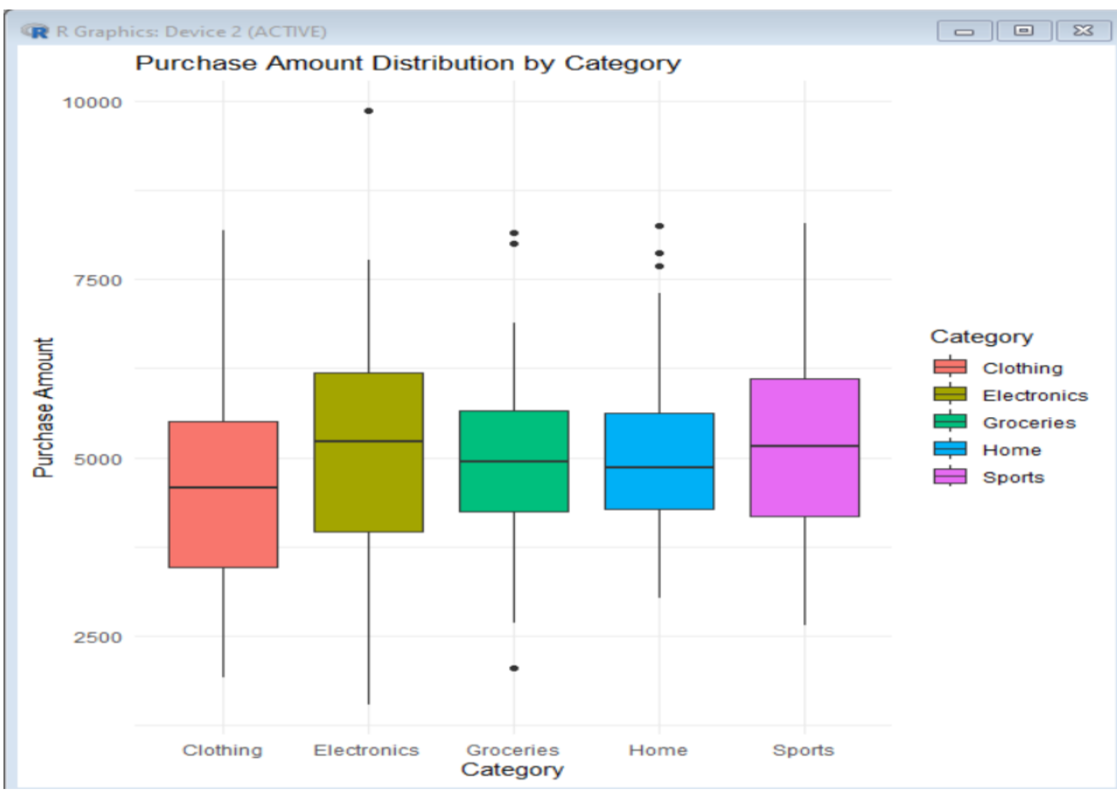
```r
# Remove negative values
customers$PurchaseAmount[customers$PurchaseAmount < 500] <- 500

# -----------------------------
# 1. Histogram
# -----------------------------
ggplot(customers, aes(x = PurchaseAmount)) +
  geom_histogram(binwidth = 500,
                 fill = "skyblue",
                 color = "black") +
  labs(title = "Distribution of Purchase Amount",
       x = "Purchase Amount",
       y = "Frequency") +
  theme_minimal()

# -----------------------------
# 2. Violin Plot
# -----------------------------
ggplot(customers,
       aes(x = Category,
           y = PurchaseAmount,
           fill = Category)) +
  geom_violin() +
  labs(title = "Purchase Amount by Category",
       x = "Category",
       y = "Purchase Amount") +
  theme_minimal()

# -----------------------------
# 3. Box Plot
# -----------------------------
ggplot(customers,
       aes(x = Category,
           y = PurchaseAmount,
           fill = Category)) +
  geom_boxplot() +
  labs(title = "Purchase Amount Distribution by Category",
       x = "Category",
       y = "Purchase Amount") +
  theme_minimal()

# -----------------------------
# 4. Density Plot
# -----------------------------
```

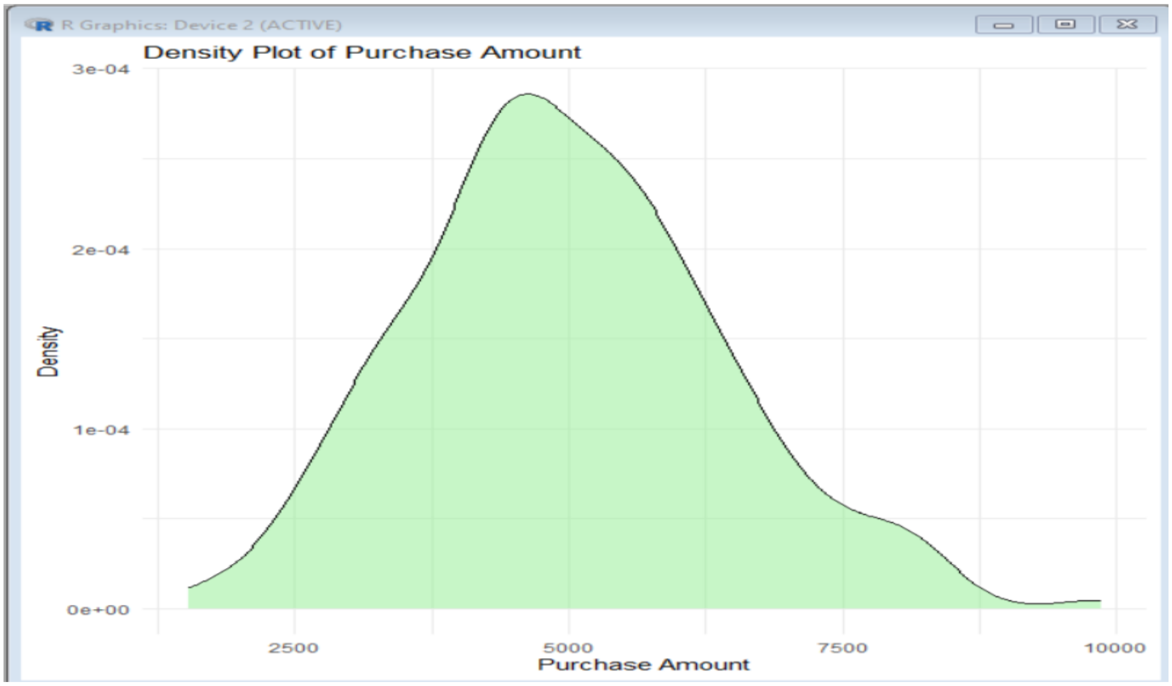
```
ggplot(customers,
       aes(x = PurchaseAmount)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(title = "Density Plot of Purchase Amount",
       x = "Purchase Amount",
       y = "Density") +
  theme_minimal()

# -----------------------------
# 5. Bar Chart
# -----------------------------
category_sales <- customers %>%
  group_by(Category) %>%
  summarise(TotalSales = sum(PurchaseAmount))

ggplot(category_sales,
       aes(x = Category,
           y = TotalSales,
           fill = Category)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Sales by Category",
       x = "Category",
       y = "Total Sales") +
  theme_minimal()
```

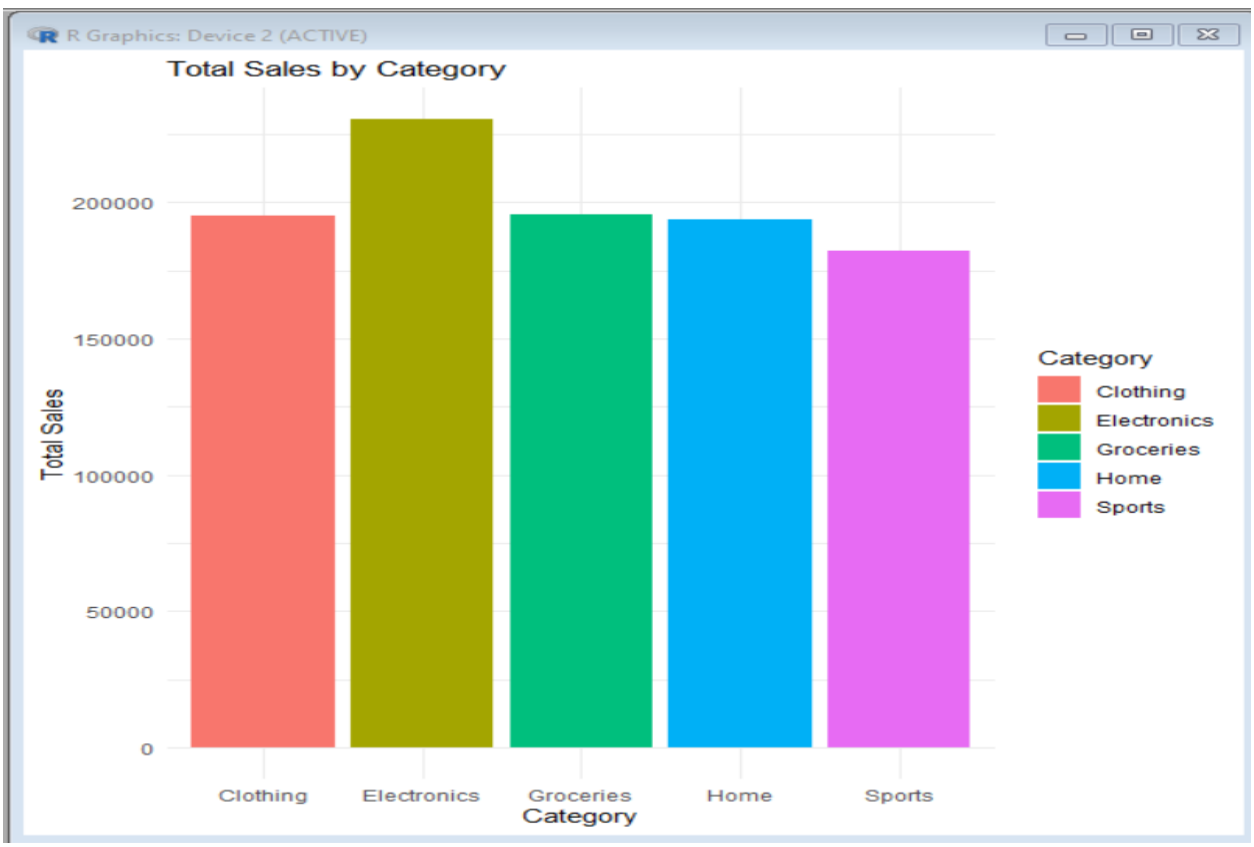
Output:




R Graphics: Device 2 (ACTIVE)
Purchase Amount by Category
10000
7500
5000
2500
Purchase Amount
Clothing
Electronics
Groceries
Home
Sports
Category
Category
Clothing
Electronics
Groceries
Home
Sports


R Graphics: Device 2 (ACTIVE)
Purchase Amount Distribution by Category
10000
7500
5000
2500
Purchase Amount
Clothing
Electronics
Groceries
Home
Sports
Category
Category
Clothing
Electronics
Groceries
Home
Sports


R Graphics: Device 2 (ACTIVE)
Density Plot of Purchase Amount
3e-04
2e-04
1e-04
0e+00
Density
2500
5000
7500
10000
Purchase Amount


R Graphics: Device 2 (ACTIVE)
Total Sales by Category
200000
150000
100000
50000
0
Total Sales
Clothing
Electronics
Groceries
Home
Sports
Category
Category
Clothing
Electronics
Groceries
Home
Sports

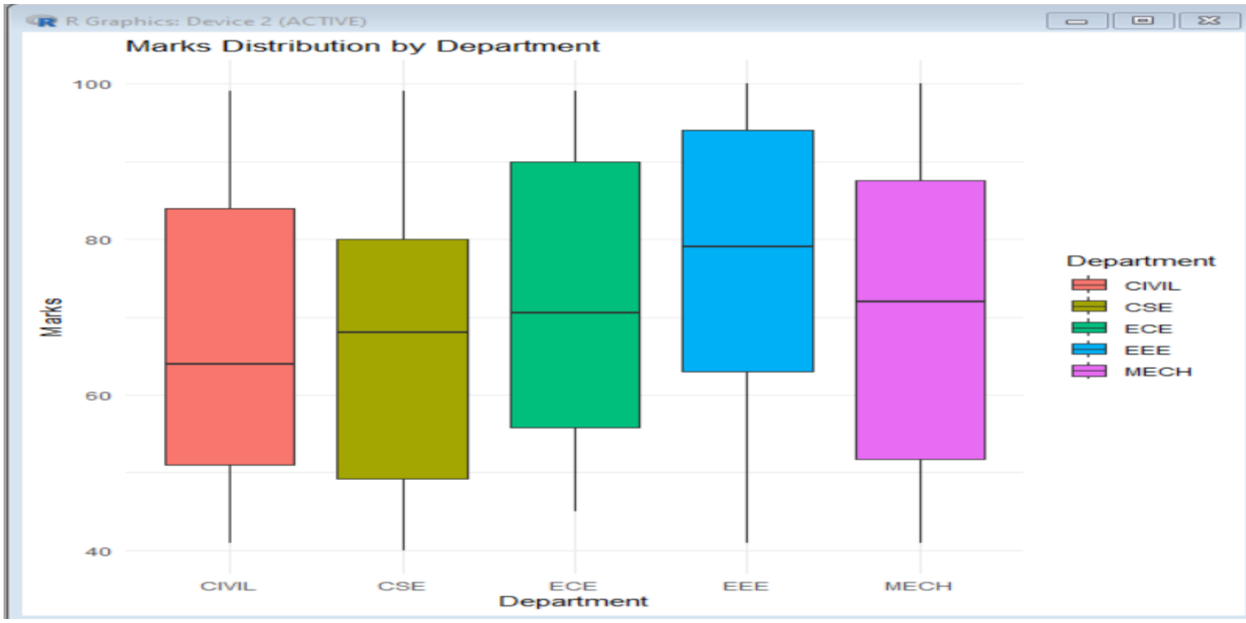
3) University Examination Performance Dashboard

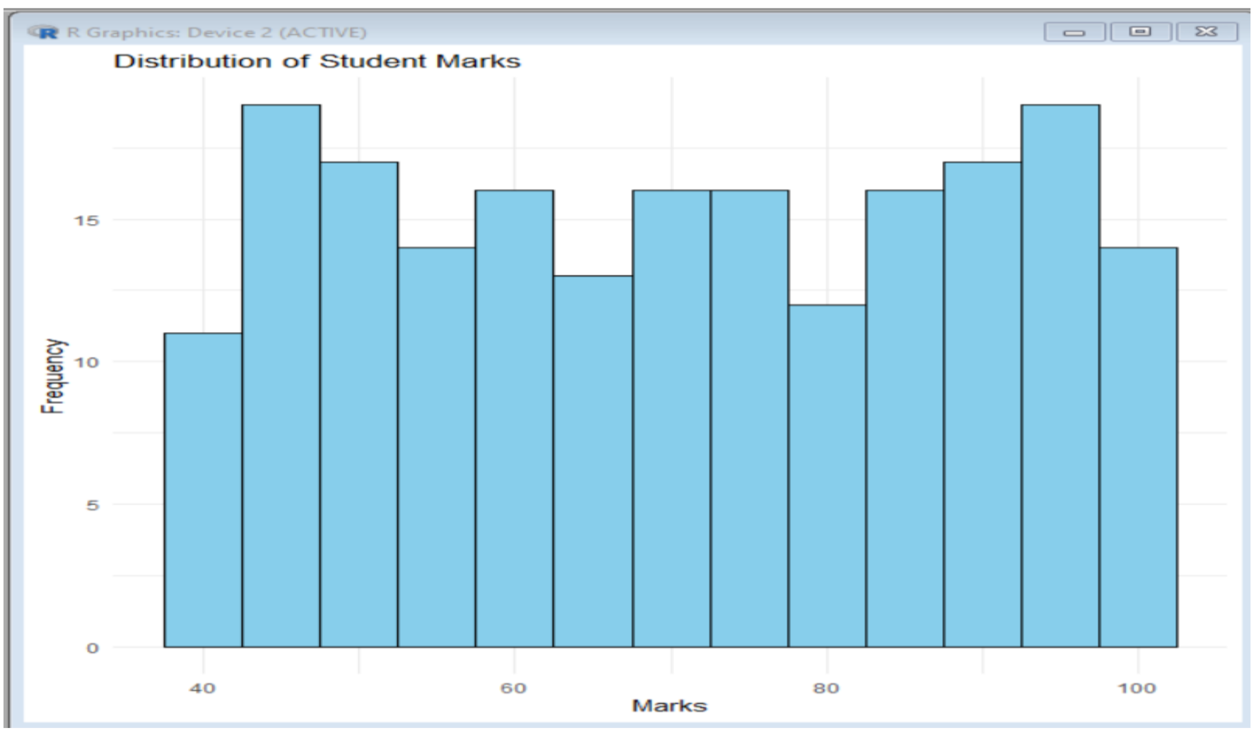
R code:

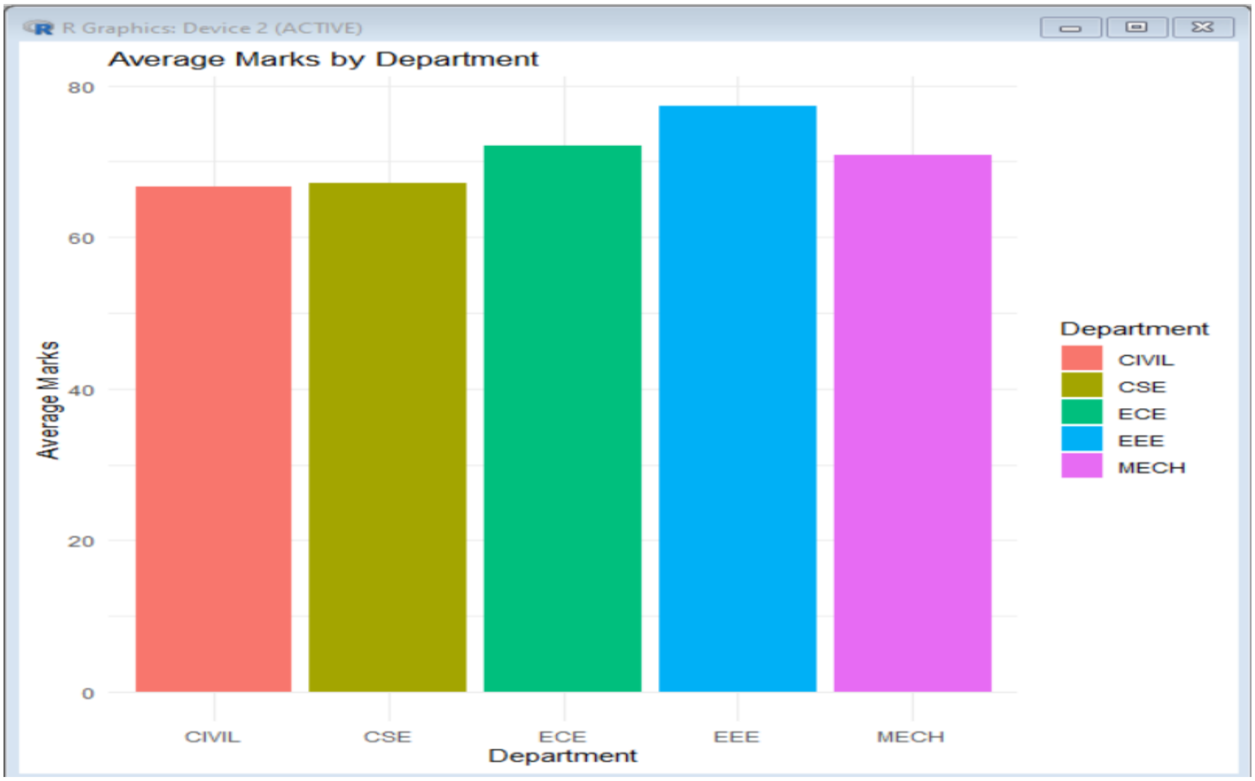
```
# Install packages (Run once)
install.packages(c("ggplot2","plotly","dplyr"))

# Load libraries
library(ggplot2)
library(plotly)
library(dplyr)

# -----------------------------
# Create Sample Dataset
# -----------------------------
set.seed(123)

students <- data.frame(
  StudentID = 1:200,
  Department = sample(c("CSE","ECE","EEE","MECH","CIVIL"),
                    200, replace = TRUE),
  Class = sample(c("A","B","C"), 200, replace = TRUE),
  Marks = sample(40:100, 200, replace = TRUE)
)

# -----------------------------
# 1. Box Plot
# -----------------------------
ggplot(students,
       aes(x = Department,
           y = Marks,
           fill = Department)) +
  geom_boxplot() +
  labs(title = "Marks Distribution by Department",
       x = "Department",
       y = "Marks") +
  theme_minimal()

# -----------------------------
# 2. Histogram
# -----------------------------
ggplot(students,
       aes(x = Marks)) +
  geom_histogram(binwidth = 5,
```

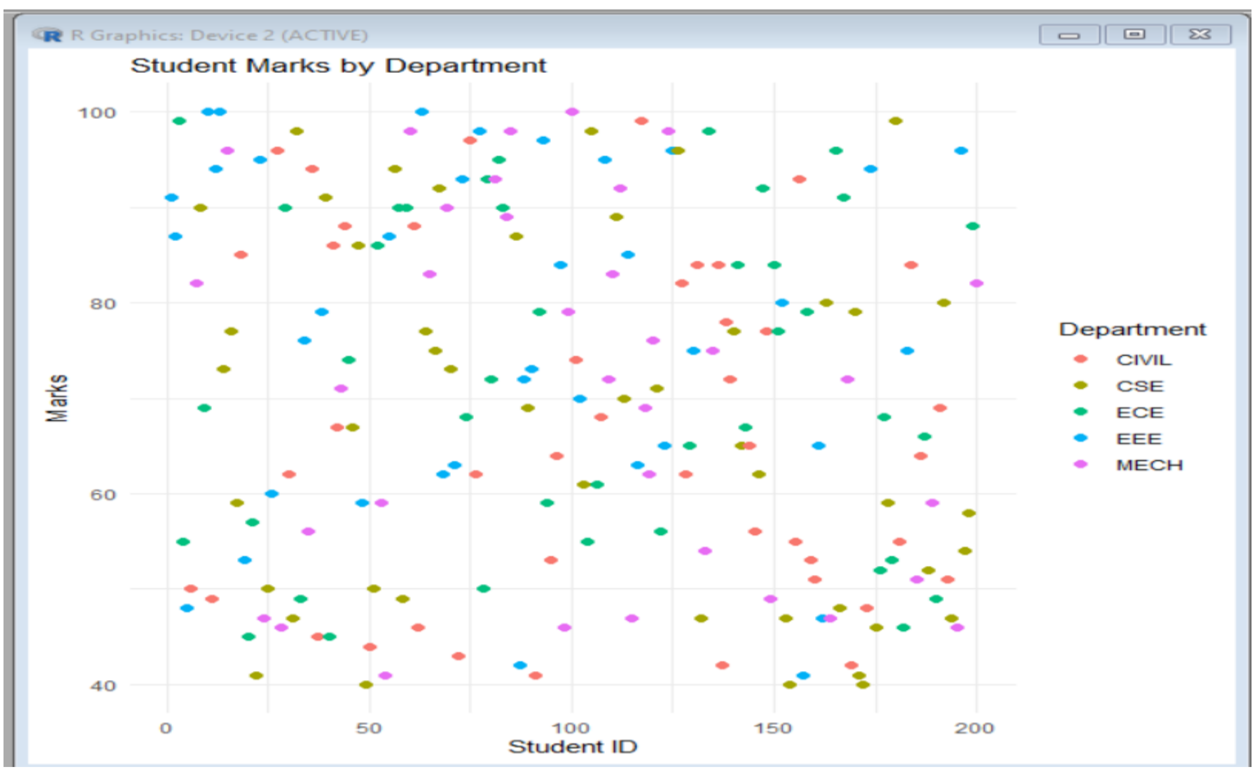
```r
             fill = "skyblue",
             color = "black") +
  labs(title = "Distribution of Student Marks",
       x = "Marks",
       y = "Frequency") +
  theme_minimal()

# -----------------------------
# 3. Bar Chart (Average Marks)
# -----------------------------
avg_marks <- students %>%
  group_by(Department) %>%
  summarise(AverageMarks = mean(Marks))

ggplot(avg_marks,
       aes(x = Department,
           y = AverageMarks,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Marks by Department",
       x = "Department",
       y = "Average Marks") +
  theme_minimal()

# -----------------------------
# 4. Scatter Plot
# -----------------------------
ggplot(students,
       aes(x = StudentID,
           y = Marks,
           color = Department)) +
  geom_point(size = 2) +
  labs(title = "Student Marks by Department",
       x = "Student ID",
       y = "Marks") +
  theme_minimal()

# -----------------------------
# 5. Interactive Plotly Chart
# -----------------------------
plot_ly(students,
        x = ~Department,
        y = ~Marks,
        color = ~Department,
```

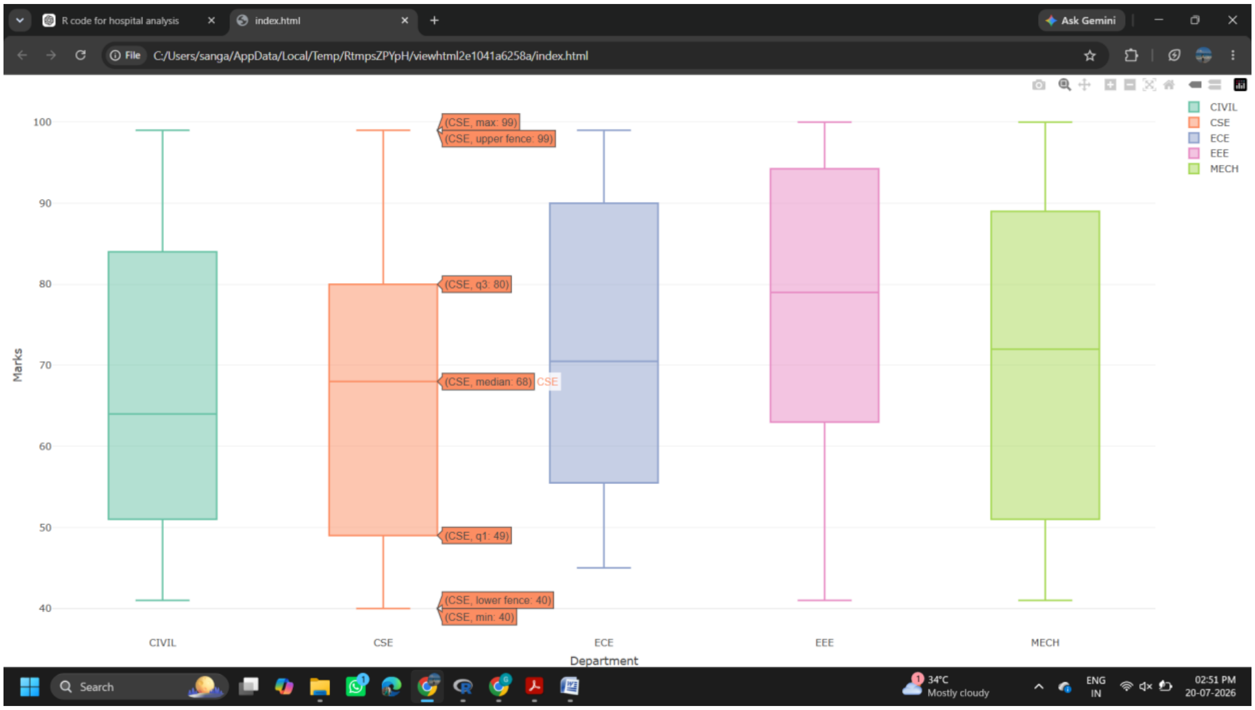
type = "box")
Output:






R Graphics: Device 2 (ACTIVE)
Average Marks by Department
80
60
40
20
0
Average Marks
CIVIL
CSE
ECE
EEE
MECH
Department
Department
CIVIL
CSE
ECE
EEE
MECH


R Graphics: Device 2 (ACTIVE)
Student Marks by Department
100
80
60
40
Marks
0
50
100
150
200
Student ID
Department
CIVIL
CSE
ECE
EEE
MECH



4) Climate and Rainfall Distribution Study:

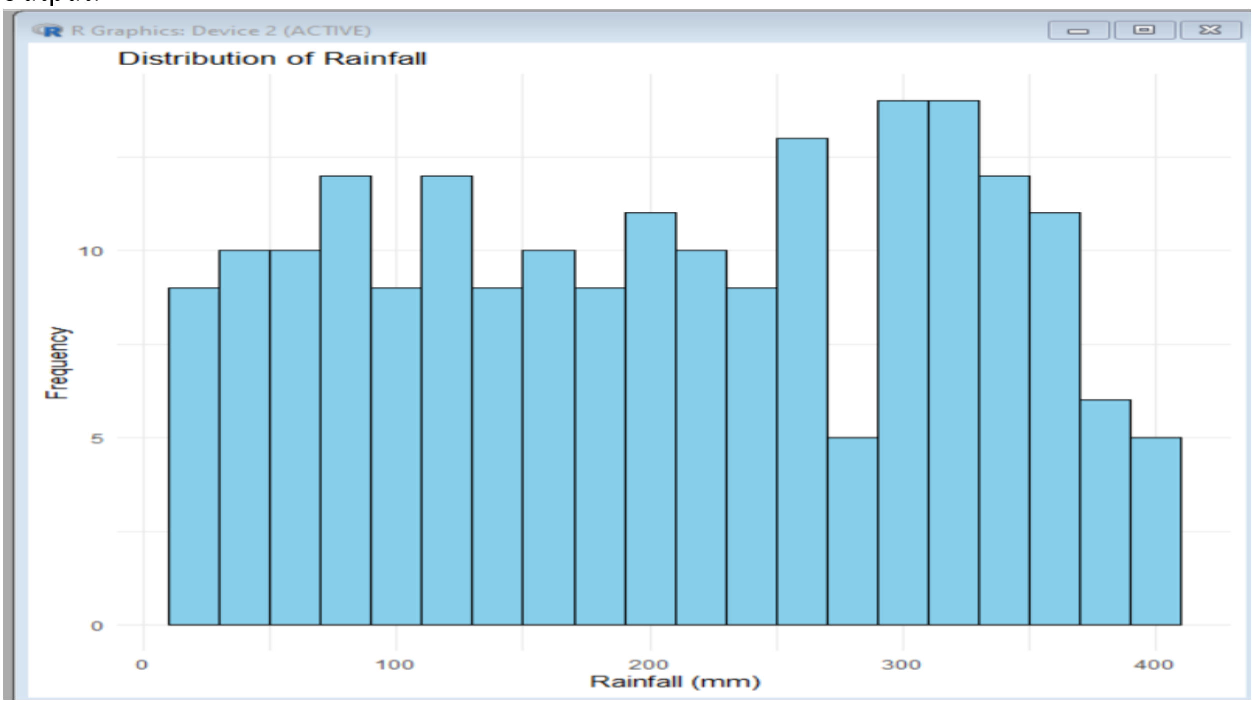
R code:

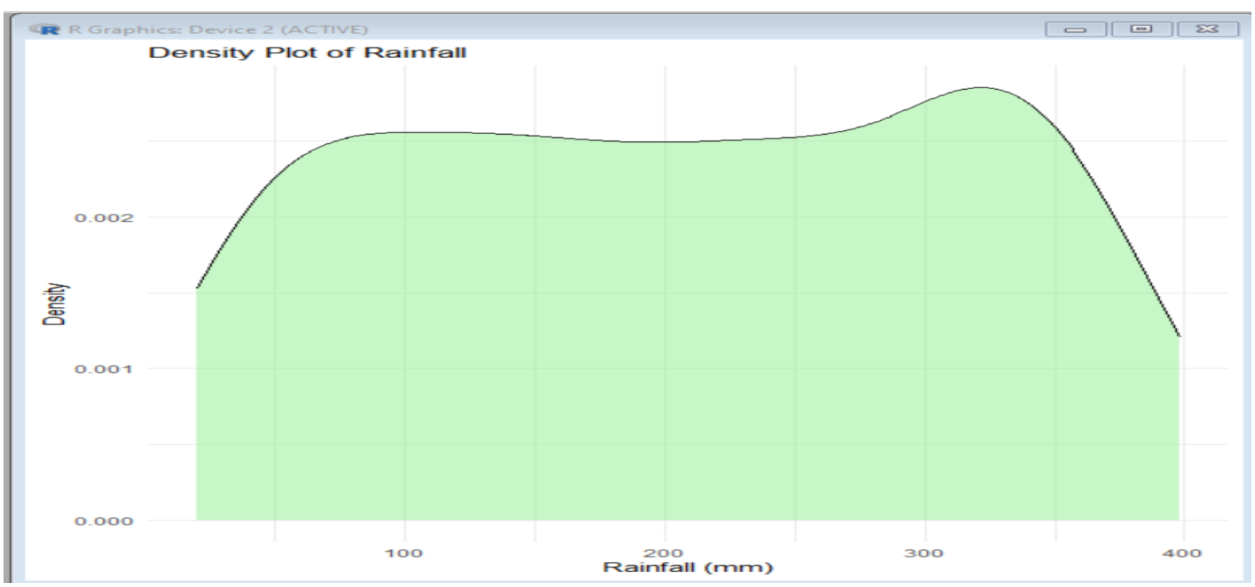
```
# Install packages (Run once)
install.packages(c("ggplot2","dplyr"))

# Load libraries
library(ggplot2)
library(dplyr)

# -----------------------------
# Create Sample Dataset
# -----------------------------
set.seed(123)

climate <- data.frame(
  City = sample(c("Chennai","Hyderabad","Bangalore","Mumbai","Delhi"),
               200, replace = TRUE),
  Month = sample(month.abb, 200, replace = TRUE),
  Rainfall = round(runif(200, 20, 400), 1),
  Temperature = round(runif(200, 18, 42), 1),
  Humidity = round(runif(200, 40, 95), 1)
)
```

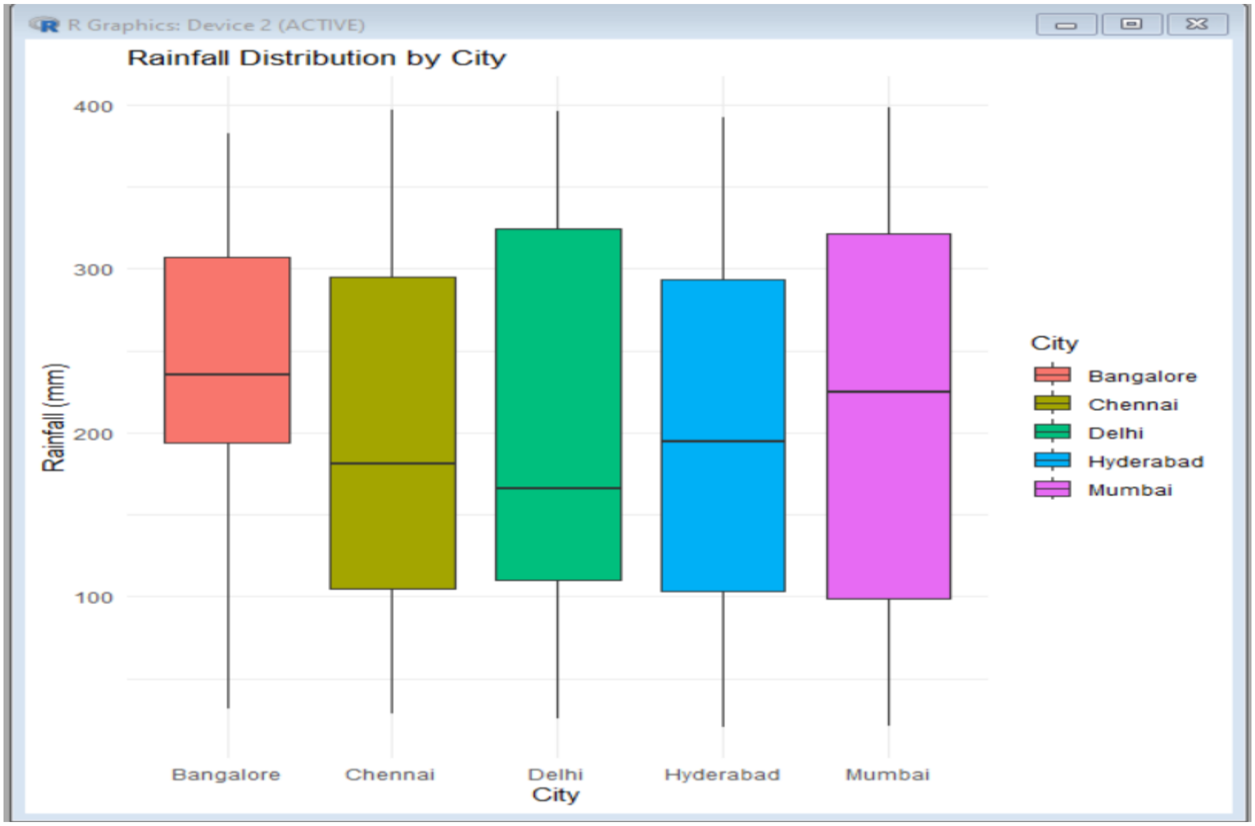
```
# -----------------------------
# 1. Histogram
# -----------------------------
ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 20,
                 fill = "skyblue",
                 color = "black") +
  labs(title = "Distribution of Rainfall",
       x = "Rainfall (mm)",
       y = "Frequency") +
  theme_minimal()

# -----------------------------
# 2. Density Plot
# -----------------------------
ggplot(climate, aes(x = Rainfall)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(title = "Density Plot of Rainfall",
       x = "Rainfall (mm)",
       y = "Density") +
  theme_minimal()

# -----------------------------
# 3. Box Plot
# -----------------------------
ggplot(climate,
       aes(x = City,
           y = Rainfall,
           fill = City)) +
  geom_boxplot() +
  labs(title = "Rainfall Distribution by City",
       x = "City",
       y = "Rainfall (mm)") +
  theme_minimal()

# -----------------------------
# 4. Bar Chart
# -----------------------------
avg_rainfall <- climate %>%
  group_by(City) %>%
  summarise(AverageRainfall = mean(Rainfall))

ggplot(avg_rainfall,
```

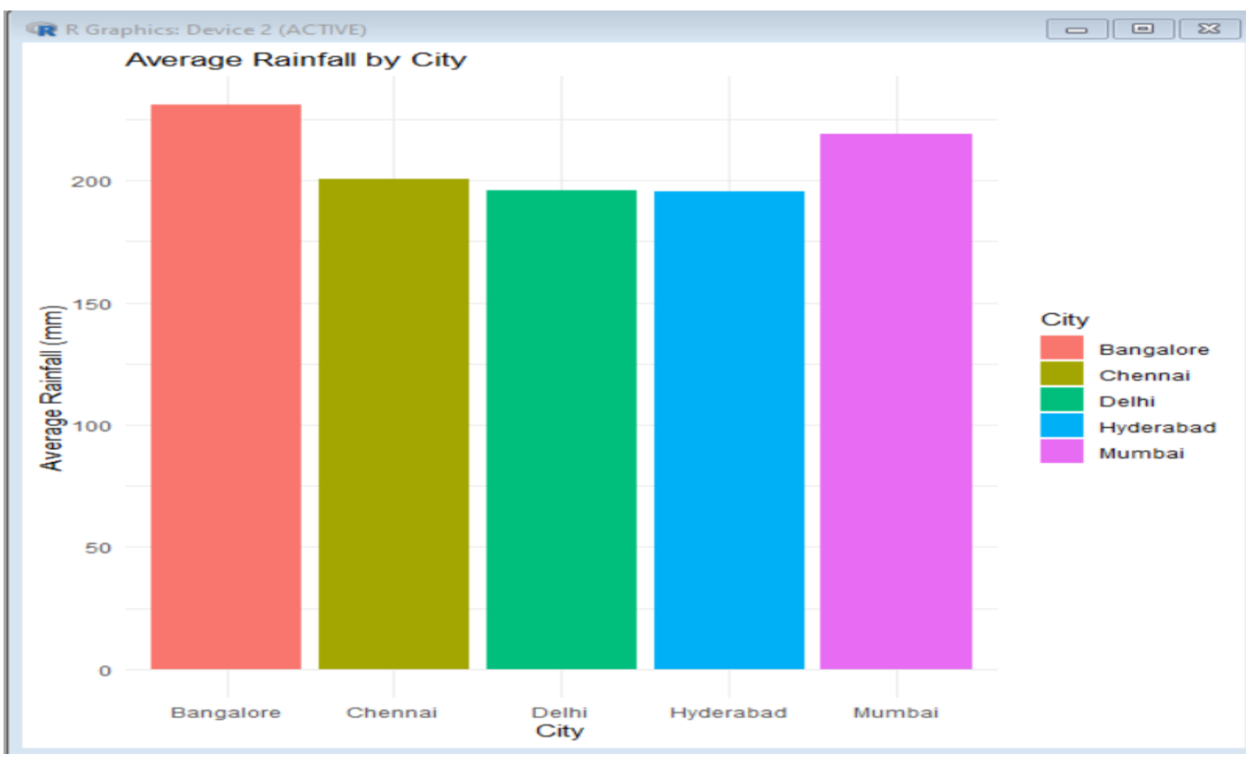
```
    aes(x = City,
        y = AverageRainfall,
        fill = City)) +
geom_bar(stat = "identity") +
labs(title = "Average Rainfall by City",
    x = "City",
    y = "Average Rainfall (mm)") +
theme_minimal()
```

Output:






R Graphics: Device 2 (ACTIVE)
Rainfall Distribution by City
400
300
200
100
Rainfall (mm)
City
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
City


R Graphics: Device 2 (ACTIVE)
Average Rainfall by City
200
150
100
50
0
Average Rainfall (mm)
City
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
City

5) Bank Loan Risk Assessment

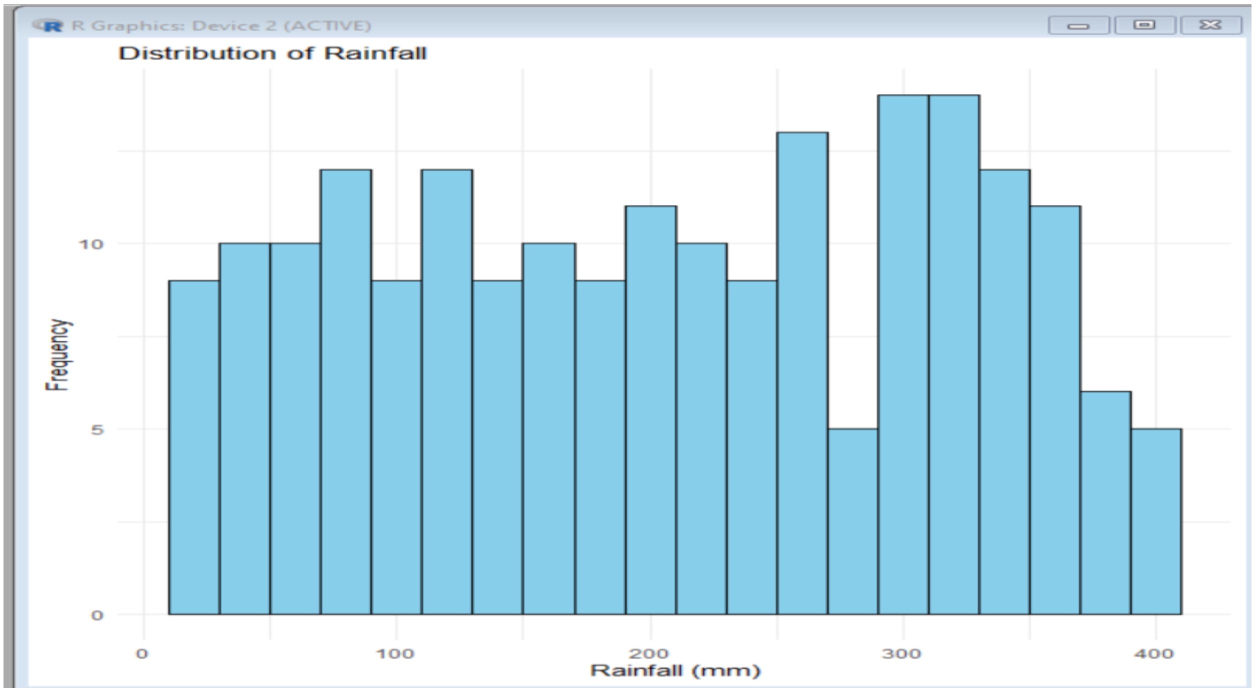
R code:

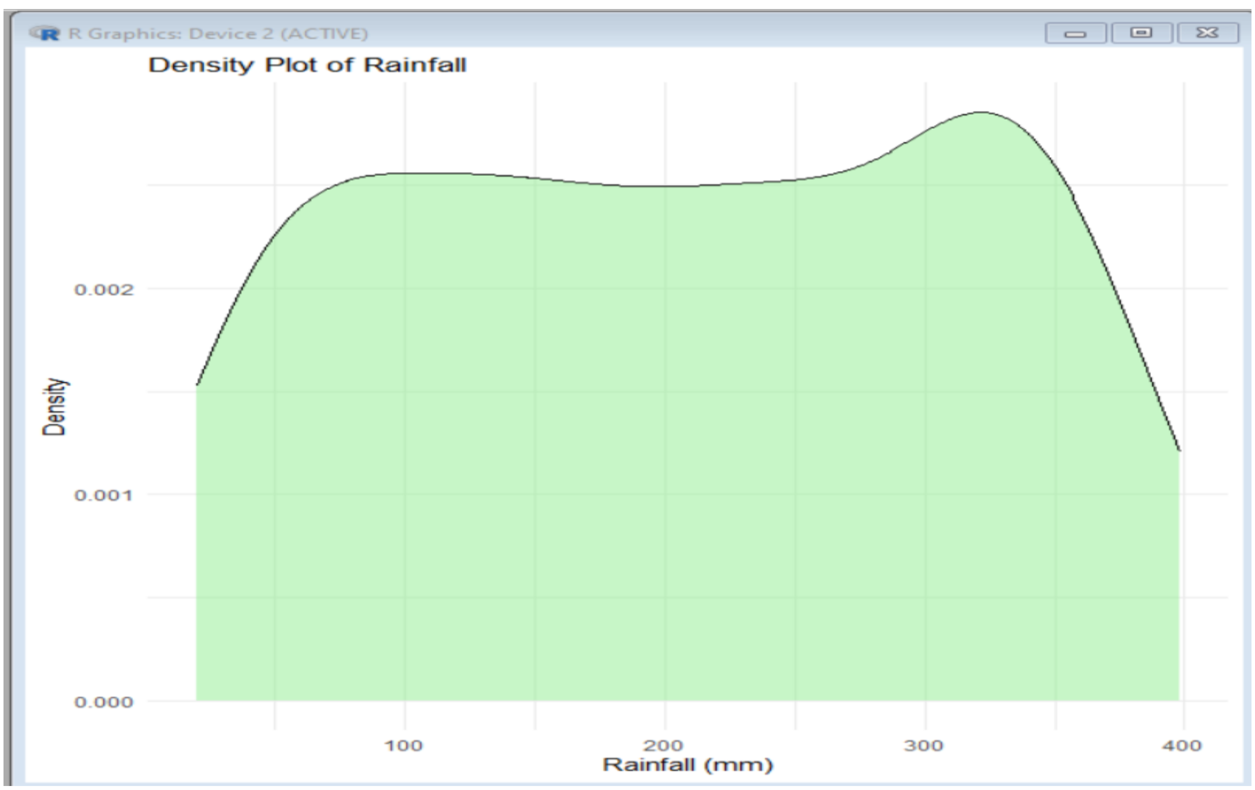
```
# Install packages (Run once)
install.packages(c("ggplot2","dplyr"))

# Load libraries
library(ggplot2)
library(dplyr)

# -----------------------------
# Create Sample Dataset
# -----------------------------
set.seed(123)

loan <- data.frame(
  CustomerID = 1:200,
  LoanAmount = round(runif(200, 50000, 1000000)),
  Income = round(runif(200, 20000, 150000)),
  CreditScore = sample(300:900, 200, replace = TRUE),
  RepaymentStatus = sample(c("Good","Average","Poor"),
                           200, replace = TRUE)
)

# -----------------------------
# 1. Histogram
# -----------------------------
ggplot(loan, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 50000,
                 fill = "skyblue",
                 color = "black") +
  labs(title = "Distribution of Loan Amount",
       x = "Loan Amount",
       y = "Frequency") +
  theme_minimal()

# -----------------------------
# 2. Box Plot
# -----------------------------
ggplot(loan,
       aes(x = RepaymentStatus,
           y = LoanAmount,
           fill = RepaymentStatus)) +
  geom_boxplot() +
```

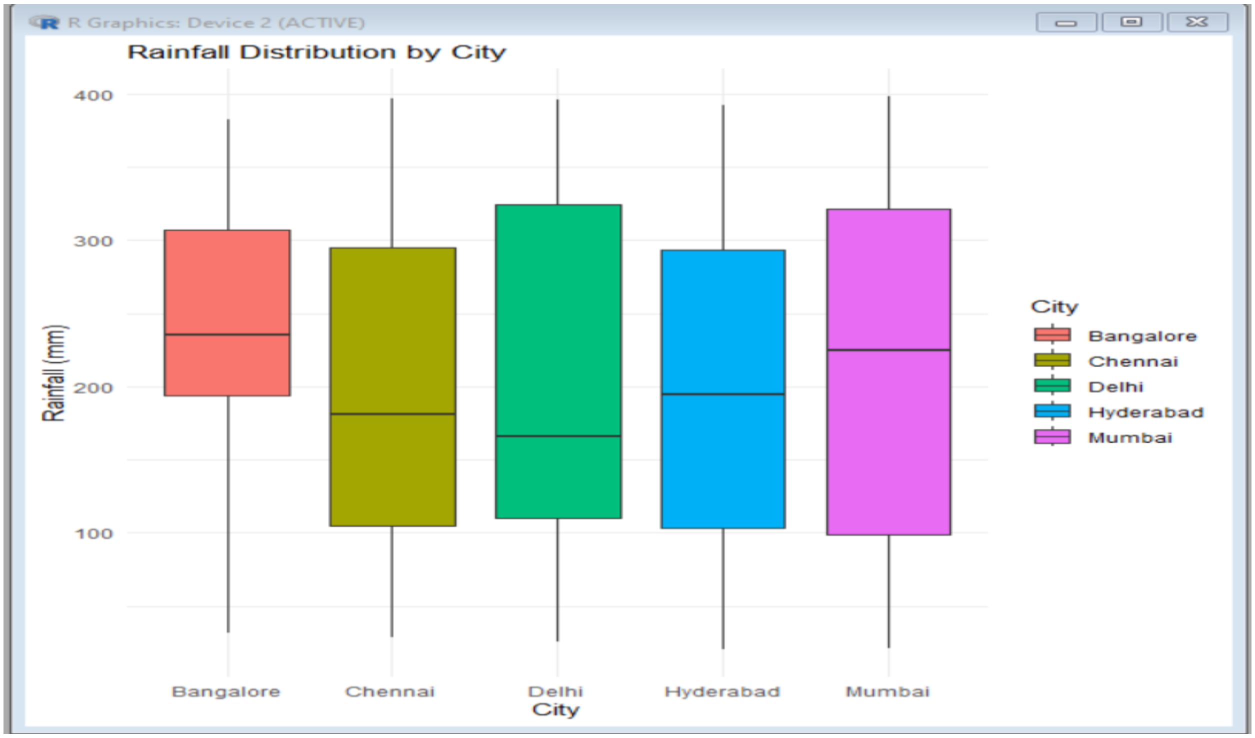
```
  labs(title = "Loan Amount by Repayment Status",
       x = "Repayment Status",
       y = "Loan Amount") +
  theme_minimal()

# ----------------------------
# 3. Scatter Plot
# ----------------------------
ggplot(loan,
       aes(x = CreditScore,
           y = LoanAmount,
           color = RepaymentStatus)) +
  geom_point(size = 2) +
  labs(title = "Loan Amount vs Credit Score",
       x = "Credit Score",
       y = "Loan Amount") +
  theme_minimal()

# ----------------------------
# 4. Bar Chart
# ----------------------------
status_count <- loan %>%
  group_by(RepaymentStatus) %>%
  summarise(Customers = n())

ggplot(status_count,
       aes(x = RepaymentStatus,
           y = Customers,
           fill = RepaymentStatus)) +
  geom_bar(stat = "identity") +
  labs(title = "Customers by Repayment Status",
       x = "Repayment Status",
       y = "Number of Customers") +
  theme_minimal()

# ----------------------------
# 5. Density Plot
# ----------------------------
ggplot(loan, aes(x = LoanAmount)) +
  geom_density(fill = "lightgreen",
               alpha = 0.5) +
  labs(title = "Density Plot of Loan Amount",
       x = "Loan Amount",
       y = "Density") +
```

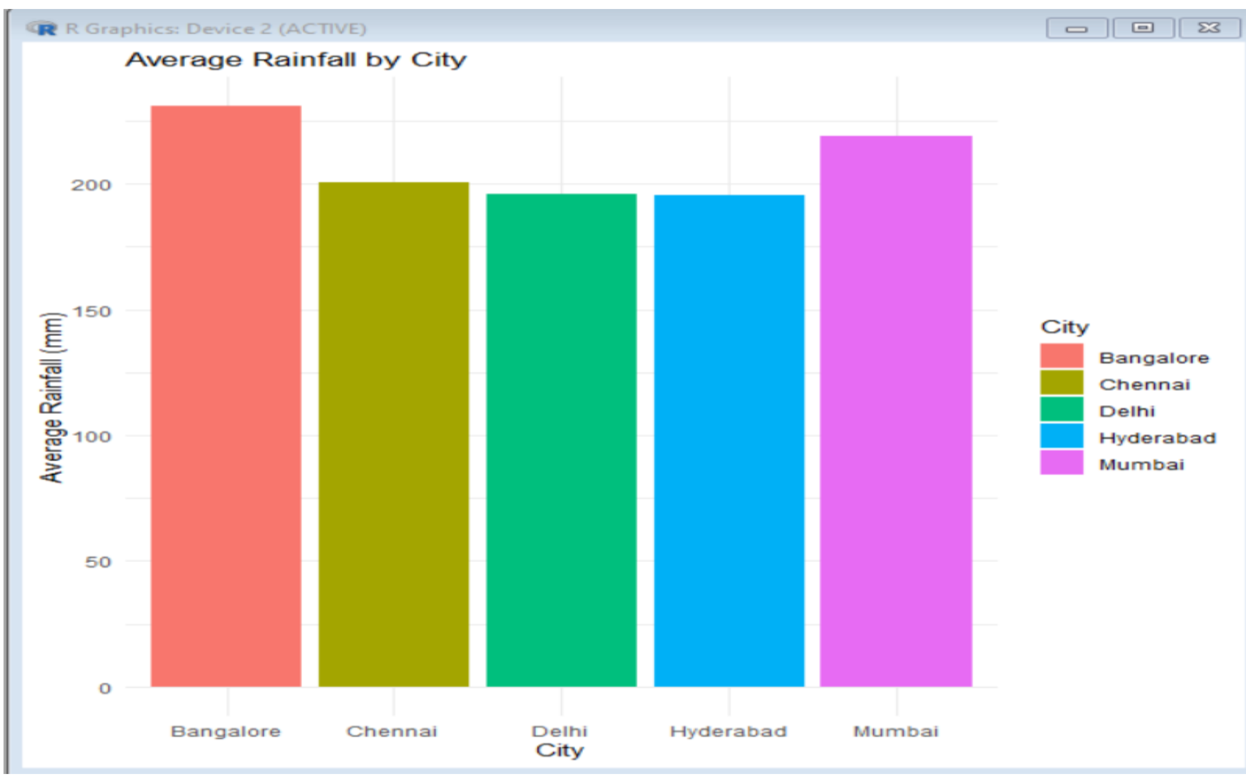
```
theme_minimal()
```

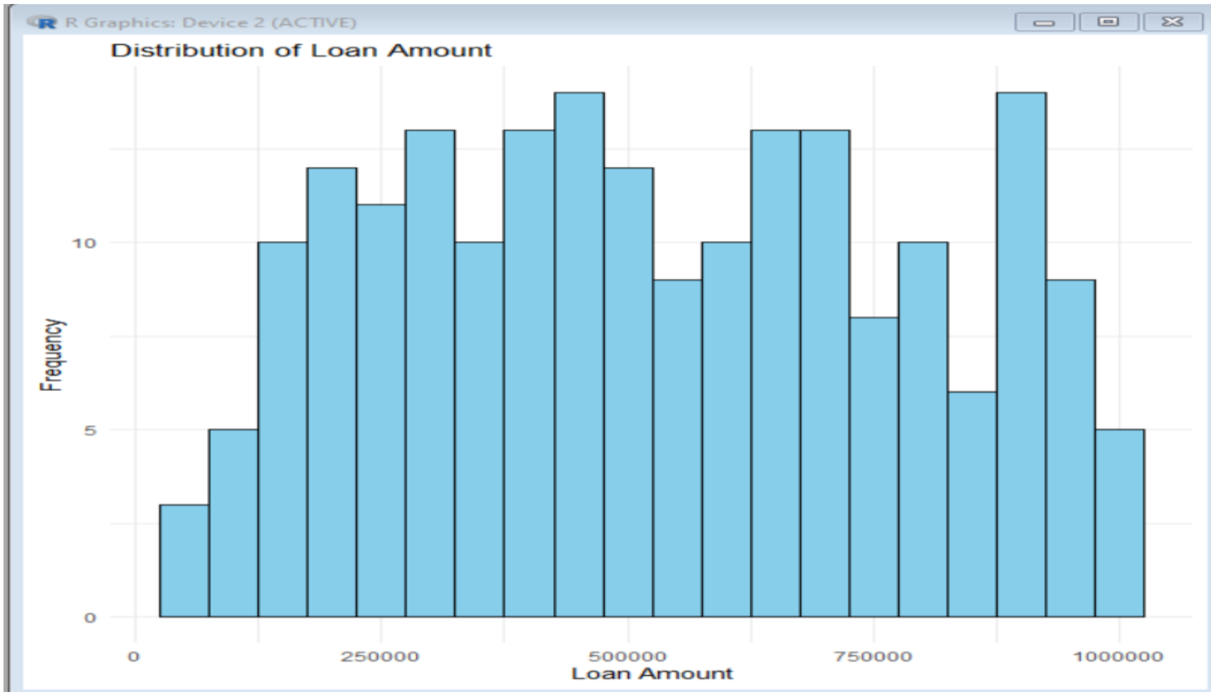
Output:






R Graphics: Device 2 (ACTIVE)
Rainfall Distribution by City
400
300
200
100
Rainfall (mm)
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
City
City
Bangalore
Chennai
Delhi
Hyderabad
Mumbai


R Graphics: Device 2 (ACTIVE)
Average Rainfall by City
200
150
100
50
0
Average Rainfall (mm)
Bangalore
Chennai
Delhi
Hyderabad
Mumbai
City
City
Bangalore
Chennai
Delhi
Hyderabad
Mumbai


R Graphics: Device 2 (ACTIVE)
Distribution of Loan Amount
Frequency
10
5
0
0
250000
500000
750000
1000000
Loan Amount